Answers learning task 18

2026 – 2027

Exercise 9.96: Consider the following frequency table of observations on the random variable X.

Values	0	1	2	3	4
Observed frequency	24	30	31	11	4

(a) Based on these 100 observations, is a Poisson distribution with a mean of 1.5 an appropriate model? Perform a goodness-of-fit procedure with $\alpha = 0.05$.

(b) Calculate the p-value for this test.

Answer

1. Notice that in this exercise, the value of the Poisson parameter $\mu = 1.5$ is given, so we do not need to estimate it. The hypotheses in this test are as follows:

   $H_0 : X$ follows a Poisson distribution with mean $\mu = 1.5$.

   $H_1 : X$ does not follow a Poisson distribution with mean $\mu = 1.5$.

   We calculate the expected frequencies of the value 0, 1, 2, 3 and 4 using the poissonpdf function:

Categorie (i)	**Observed (O_i)**	**Expected (E_i)**
0	24	$100 \cdot \text{poissonpdf}(\mu = 1.5, k = 0) \approx 22.3130$
1	30	$100 \cdot \text{poissonpdf}(\mu = 1.5, k = 1) \approx 33.4695$
2	31	$100 \cdot \text{poissonpdf}(\mu = 1.5, k = 2) \approx 25.1021$
3	11	$100 \cdot \text{poissonpdf}(\mu = 1.5, k = 3) \approx 12.5511$
≥ 4	4	$100 \cdot (1 - \text{poissoncdf}(\mu = 1.5, k = 3)) \approx 6.5642$

   The χ^2 test statistic can then be computed as follows:

$$\begin{aligned}\chi^2 &= \frac{(O_0 - E_0)^2}{E_0} + \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \frac{(O_3 - E_3)^2}{E_3} + \frac{(O_{\geq 4} - E_{\geq 4})^2}{E_{\geq 4}} \\ &= \frac{(24 - 22.3130)^2}{22.3130} + \ldots + \frac{(4 - 6.5642)^2}{6.5642} \\ &\approx 3.0663.\end{aligned}$$

   Under assumption of H_0, the test statistic comes from a chi-square distribution with $\text{df} = n - p - 1 = 5 - 0 - 1 = 4$ degrees of freedom, where n is the number of categories and p the number of estimated parameters. The critical region ($\alpha = 0.05$, χ^2-test) is $Z = [C_R, \infty)$, and we can determine the value of C_R using the "numeric solver":

$$\begin{aligned}y_1 &= \chi^2\text{cdf}(\text{lower} = X, \text{upper} = 10^{99}, \text{df} = 5 - 0 - 1 = 4), \\ y_2 &= \alpha = 0.05.\end{aligned}$$

The numeric solver returns a value $C_R \approx 9.4877$. As $\chi^2 = 3.0663$ does not lie in the critical region, we fail to reject H_0.

2. The p-value corresponding to $\chi^2 = 3.0663$ is

$$p = P(X^2 \geq 3.0663) = \chi^2\text{cdf}(\text{lower} = 3.0663, \text{upper} = 10^{99}, \text{df} = n - p - 1 = 4) \approx 0.5468.$$

As $p > \alpha = 0.05$, we again fail to reject H_0. There is not enough evidence to reject the claim that the data comes from a Poisson distribution with mean $\mu = 1.5$.

Exercise 9.97: Let X denote the number of flaws observed on a large coil of galvanized steel. Of 80 coils inspected, the following data were observed for the values of X:

Values	1	2	3	4	5	6	7	8
Observed frequency	1	11	8	13	11	12	10	9

(a) Does the assumption of the Poisson distribution seem appropriate as a probability model for these data? Use $\alpha = 0.01$.

(b) Calculate the p-value for this test.

Answer

Note: 75 coils

(a) $\chi^2 = 6.4741 \notin [16.81, \infty) \implies$ don't reject H_0

(b) $p = 0.3722 > \alpha = 0.01 \implies$ don't reject H_0

Exercise 9.99: Consider the following frequency table of observations on the random variable X:

Values	0	1	2	3	4
Frequency	4	14	10	20	2

(a) Based on these 50 observations, is a binomial distribution with $n = 6$ and $p = 0.25$ an appropriate model? Perform a goodness-of-fit procedure with $\alpha = 0.05$.

Answer

(a) $\chi^2 = 32.3778 \in [9.49, \infty) \implies$ reject H_0

(b) $p = 0.000002 \leq \alpha = 0.05 \implies$ reject H_0

Exercise 9.144: Consider the spot weld shear strength data in the following table:

5409	5431	5475	5442	5376	5388	5459	5422	5416	5435
5420	5430	5401	5446	5487	5416	5382	5357	5388	5457
5407	5469	5415	5377	5454	5375	5409	5459	5445	5429
5463	5408	5481	5452	5422	5354	5421	5406	5444	5466
5399	5391	5477	5447	5330	5473	5423	5441	5412	5384
5445	5436	5454	5453	5428	5419	5465	5427	5421	5396
5381	5425	5388	5388	5378	5481	5388	5440	5482	5406
5401	5411	5399	5431	5440	5413	5406	5341	5452	5420
5458	5485	5431	5416	5431	5390	5399	5435	5386	5462
5383	5401	5407	5385	5440	5422	5448	5366	5430	5417

Does the normal distribution seem to be a reasonable model for these data? Perform an appropriate goodness-of-fit test to answer this question.

Answer

5 cells with equal expected frequencies ($= 20$): $\chi^2 = 0.8 \notin [5.99, \infty) \implies p = 0.6703 > \alpha = 0.05$, don't reject H_0

10 cells with equal expected frequencies ($= 10$): $\chi^2 = 3.2 \notin [14.07, \infty) \implies p = 0.8659 > \alpha = 0.05$, don't reject H_0

A normal distribution for this model seems reasonable.